

# Commerce XI Maths App

## Exercise 2.1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 8

## Question 1 - Q1

Find range of the following data:

575, 609, 335, 280, 729, 544, 852, 427, 967, 250

**Correct answer: 717**

Explanation and steps

Use range = largest value - smallest value. Substitution gives 717.

## Question 2 - Q2

The following data gives number of typing mistakes done by Radha during a week.

Find the range of the data.

| Day             | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday |
|-----------------|--------|---------|-----------|----------|--------|----------|
| No. of mistakes | 15     | 20      | 21        | 12       | 17     | 10       |

**Correct answer: 11**

Explanation and steps

Use range = largest value - smallest value. Substitution gives 11.

## Question 3 - Q3

Find range for the following data.

| Classes   | 62-64 | 64-66 | 66-68 | 68-70 | 70-72 |
|-----------|-------|-------|-------|-------|-------|
| Frequency | 5     | 3     | 4     | 5     | 3     |

**Correct answer: 10**

Explanation and steps

Use range = largest value - smallest value. Substitution gives 10.

## Question 4 - Q4

Find Q.D. for the following data:

3, 16, 8, 15, 19, 11, 5, 17, 9, 5, 3.

**Correct answer: 5.5**

Explanation and steps

Locate Q1 and Q3, then use  $Q.D. = (Q3 - Q1)/2$ . The verified result is 5.5.

## Question 5 - Q5

Given below are the prices of shares of a company for the last 10 days.

172, 164, 188, 214, 190, 237, 200, 195, 208, 230

Find Q.D.

**Correct answer: 17**

Explanation and steps

Locate Q1 and Q3, then use  $Q.D. = (Q3 - Q1)/2$ . The verified result is 17.

## Question 6 - Q6

Calculate Q.D. for the following data.

| X | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
|---|----|----|----|----|----|----|----|
| F | 6  | 5  | 3  | 2  | 4  | 7  | 3  |

**Correct answer: 2**

Explanation and steps

Locate Q1 and Q3, then use  $Q.D. = (Q3 - Q1)/2$ . The verified result is 2.

## Question 7 - Q7

Following data gives the age distribution of 250 employees of a firm.

Calculate Q.D. of the distribution.

| Age (in years)   | 20-25 | 25-30 | 30-35 | 35-40 | 40-45 | 45-50 |
|------------------|-------|-------|-------|-------|-------|-------|
| No. of employees | 30    | 40    | 60    | 50    | 46    | 14    |

**Correct answer: 5.625**

Explanation and steps

Locate Q1 and Q3, then use  $Q.D. = (Q3 - Q1)/2$ . The verified result is 5.625.

## Question 8 - Q8

Following data gives the weight of boxes.

Calculate Q.D. for the data.

| Weight (kg)  | 10-12 | 12-14 | 14-16 | 16-18 | 18-20 | 20-22 |
|--------------|-------|-------|-------|-------|-------|-------|
| No. of boxes | 3     | 7     | 16    | 14    | 18    | 2     |

**Correct answer: 1.965**

Explanation and steps

Locate Q1 and Q3, then use  $Q.D. = (Q3 - Q1)/2$ . The verified result is 1.965.